

A Bath-Temperature Extension of the LTE Hammer–Rosen Marshak-Wave Model with Successive Global History Corrections

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Abstract

The original LTE Hammer–Rosen (HR) theory assumes that the surface temperature $T_s(t)$ is prescribed. Here the input is instead the incident bath temperature $T_b(t)$. The bath problem is reduced to the determination of an effective surface temperature $T_s(t)$, after which the original HR front and profile formulas are applied without modification. A compact zeroth-order fluence model is systematically improved by successive global history corrections. At every correction order the calculation requires only cumulative fluence integrals and one monotone scalar algebraic solve at each time; no derivative of $T_b(t)$ and no implicit time-marching boundary evolution are required. If the history iteration converges, it recovers the continuous HR boundary-energy closure used by the Cohen–Heizler appendix model.

1 LTE diffusion equation and material model

The LTE material model is

$$e = fT^\beta \rho^{-\mu}, \quad \frac{1}{\kappa_R} = gT^\alpha \rho^{-\lambda}, \quad (1)$$

where e is the specific material energy, κ_R is the Rosseland opacity, ρ is constant, and $f, \beta, \mu, g, \alpha, \lambda$ are material constants. The LTE diffusion equation can be written as

$$\boxed{\frac{\partial T^\beta}{\partial t} = C \frac{\partial^2 T^n}{\partial x^2}} \quad (2)$$

with

$$\boxed{n = 4 + \alpha, \quad \epsilon = \frac{\beta}{n}} \quad (3)$$

and

$$\boxed{C = \frac{16g\sigma}{3nf} \rho^{-2+\mu-\lambda}.} \quad (4)$$

Here σ is the Stefan–Boltzmann constant in the unit system used for the HR coefficients.

2 Original surface-temperature HR formulas

For a prescribed surface temperature $T_s(t)$, define

$$H(t) = T_s^n(t). \quad (5)$$

The first-order HR front formula is

$$x_F^2(t) = K_\epsilon C H^{-\epsilon}(t) \int_0^t H(t') dt' \quad (6)$$

where

$$K_\epsilon = \frac{2 + \epsilon}{1 - \epsilon}. \quad (7)$$

Equivalently, since $H^{-\epsilon} = T_s^{-\beta}$,

$$x_F^2(t) = K_\epsilon C T_s^{-\beta}(t) \int_0^t T_s^n(t') dt'. \quad (8)$$

The HR material energy per unit area is

$$E(t) = f \rho^{1-\mu} (1 - \epsilon) x_F(t) H^\epsilon(t). \quad (9)$$

Combining (6) and (9) gives

$$E^2(t) = Z_1 H^\epsilon(t) \int_0^t H(t') dt' \quad (10)$$

with

$$Z_1 = f^2 \rho^{2(1-\mu)} (2 + \epsilon)(1 - \epsilon) C. \quad (11)$$

3 Bath boundary condition and exact HR boundary relation

The bath-driven problem prescribes $T_b(t)$ rather than $T_s(t)$. We use the Marshak boundary condition

$$F(t) = 2\sigma [T_b^4(t) - T_s^4(t)]. \quad (12)$$

Energy conservation gives

$$E(t) = \int_0^t F(t') dt'. \quad (13)$$

Define

$$u(t) = \left[\frac{T_s(t)}{T_b(t)} \right]^4, \quad (14)$$

so that

$$T_s(t) = T_b(t) u^{1/4}(t), \quad H(t) = u^{n/4}(t) T_b^n(t). \quad (15)$$

Also define the two bath fluences

$$J_4(t) = \int_0^t T_b^4(t') dt', \quad J_n(t) = \int_0^t T_b^n(t') dt'. \quad (16)$$

Substituting (14)–(16) into (10)–(13) gives the continuous HR boundary integral equation

$$\begin{aligned} & \left\{ 2\sigma \left[J_4(t) - \int_0^t u(t') T_b^4(t') dt' \right] \right\}^2 \\ &= Z_1 T_b^\beta(t) u^{\beta/4}(t) \int_0^t u^{n/4}(t') T_b^n(t') dt'. \end{aligned} \quad (17)$$

The Cohen–Heizler appendix method solves the same HR boundary-energy closure by implicit time marching. The hierarchy below instead treats (17) by successive global history corrections.

4 Successive global history-correction hierarchy

4.1 Zeroth-order compact fluence model

The compact approximation replaces the past ratio by its current value inside both history integrals. The zeroth-order ratio $u^{(0)}(t)$ is therefore the unique positive solution of

$$1 - u^{(0)}(t) = \left[\frac{Z_1 T_b^\beta(t) J_n(t)}{[2\sigma J_4(t)]^2} \right]^{1/2} \left[u^{(0)}(t) \right]^{(n+\beta)/8}. \quad (18)$$

This is the original compact fluence model.

4.2 History factors and the next correction

Given the order- k approximation $u^{(k)}(t)$, define the normalized history factors

$$I_4^{(k)}(t) = \frac{\int_0^t u^{(k)}(t') T_b^4(t') dt'}{u^{(k)}(t) J_4(t)}. \quad (19)$$

$$I_n^{(k)}(t) = \frac{\int_0^t \left[u^{(k)}(t') \right]^{n/4} T_b^n(t') dt'}{\left[u^{(k)}(t) \right]^{n/4} J_n(t)}. \quad (20)$$

The zeroth-order constant-ratio approximation corresponds to $I_4 = I_n = 1$. The next history-corrected ratio $u^{(k+1)}(t)$ is the unique positive solution of

$$1 - I_4^{(k)}(t) u^{(k+1)}(t) = \left[\frac{Z_1 T_b^\beta(t) J_n(t)}{[2\sigma J_4(t)]^2} I_n^{(k)}(t) \right]^{1/2} \left[u^{(k+1)}(t) \right]^{(n+\beta)/8}. \quad (21)$$

Thus an order- K boundary model consists of the zeroth-order solve (18) followed by K applications of (19)–(21). In particular, $K = 0$ is the compact fluence model, $K = 1$ is the leading history correction, and larger K retain progressively more of the actual surface-to-bath history.

For fixed t , define the residual

$$\begin{aligned} \Phi_{k+1}(u; t) &= 1 - I_4^{(k)}(t) u \\ &\quad - \left[\frac{Z_1 T_b^\beta(t) J_n(t)}{[2\sigma J_4(t)]^2} I_n^{(k)}(t) \right]^{1/2} u^{(n+\beta)/8}. \end{aligned} \quad (22)$$

Because $\Phi_{k+1}(0; t) = 1$, $\Phi_{k+1}(1/I_4^{(k)}; t) < 0$, and $\partial\Phi_{k+1}/\partial u < 0$, equation (21) has exactly one positive root in

$$0 < u^{(k+1)}(t) < \frac{1}{I_4^{(k)}(t)}. \quad (23)$$

The upper bound need not equal one; this allows $T_s(t) > T_b(t)$ during cooling.

At a converged fixed point $u^{(k+1)} = u^{(k)} = u$, the factors (19)–(20) substitute the actual histories into (21), which then reduces exactly to (17). Hence the converged global iteration and the continuous Cohen–Heizler HR boundary closure are equivalent, apart from quadrature and algebraic-solver tolerances.

5 Surface temperature and application of the original HR formulas

After the chosen correction order K has been reached, define

$$T_s^{(K)}(t) = T_b(t) \left[u^{(K)}(t) \right]^{1/4}. \quad (24)$$

The bath model ends at this point: its only purpose is to infer the effective surface history $T_s^{(K)}(t)$. The front and profile are then obtained from the original surface-temperature HR theory.

Using (8) with $T_s = T_s^{(K)}$ gives

$$x_F^2(t) = K_\epsilon C \left[T_s^{(K)}(t) \right]^{-\beta} \int_0^t \left[T_s^{(K)}(t') \right]^n dt'. \quad (25)$$

For the profile, define

$$H_K(t) = \left[T_s^{(K)}(t) \right]^n, \quad y = \frac{x}{x_F(t)}, \quad (26)$$

and

$$A_K(t) = \frac{x_F^2(t)}{C H_K^{1-\epsilon}(t)} \frac{\dot{H}_K(t)}{H_K(t)}. \quad (27)$$

The first-order HR temperature profile is

$$T(x, t) = T_s^{(K)}(t) \left[(1 - y) \left(1 + \frac{\epsilon}{2} (1 - A_K(t)) y \right) \right]^{1/[n(1-\epsilon)]} \quad (28)$$

for $0 \leq y \leq 1$, and $T(x, t) = 0$ for $y > 1$. A leading-order profile is

$$T(x, t) \approx T_s^{(K)}(t) \left(1 - \frac{x}{x_F(t)} \right)^{1/n}. \quad (29)$$

6 Numerical algorithm

Given tabulated $T_b(t)$ and a requested correction order K :

1. Interpolate $T_b(t)$ on a convenient time grid. For noisy data, linear interpolation is usually safest.
2. Compute $J_4(t_i)$ and $J_n(t_i)$ from (16) by cumulative trapezoidal integration.
3. At every time with nonzero bath and nonzero fluences, solve (18) for $u^{(0)}(t_i)$ by Brent's method or bisection on $(0, 1)$. Before the bath turns on, set $T_s = x_F = 0$.
4. For $k = 0, 1, \dots, K - 1$:
 - (a) evaluate the two cumulative history integrals in (19)–(20) by the trapezoidal rule;
 - (b) compute $I_4^{(k)}(t_i)$ and $I_n^{(k)}(t_i)$;
 - (c) solve (21) for $u^{(k+1)}(t_i)$ by Brent's method or bisection on the guaranteed interval $(0, 1/I_4^{(k)}(t_i))$.
5. Set $T_s^{(K)}(t_i)$ from (24).
6. Apply the original HR front formula (25), using a cumulative trapezoidal integral of $[T_s^{(K)}]^n$.

7. For profiles, form $H_K = [T_s^{(K)}]^n$, estimate $\dot{H}_K$ by finite differences, compute A_K from (27), and use (28).

If a fixed order is not prescribed, iterate until, for example,

$$\max_i \frac{|u^{(k+1)}(t_i) - u^{(k)}(t_i)|}{\max\{u^{(k+1)}(t_i), u_{\text{floor}}\}} < \tau, \quad (30)$$

where τ is the requested tolerance and u_{floor} prevents division by zero near turn-on. Convergence should be checked rather than inferred only from a chosen value of K .

7 Self-contained summary

The LTE diffusion problem is

$$\frac{\partial T^\beta}{\partial t} = C \frac{\partial^2 T^n}{\partial x^2}, \quad n = 4 + \alpha, \quad \epsilon = \frac{\beta}{n}, \quad (31)$$

with

$$C = \frac{16g\sigma}{3nf} \rho^{-2+\mu-\lambda}, \quad (32)$$

and material model

$$e = fT^\beta \rho^{-\mu}, \quad \frac{1}{\kappa_R} = gT^\alpha \rho^{-\lambda}. \quad (33)$$

The semi-infinite material is initially cold, hydrodynamics is neglected, LTE is assumed, and the imposed bath temperature obeys the Marshak condition

$$F(t) = 2\sigma [T_b^4(t) - T_s^4(t)]. \quad (34)$$

The bath calculation serves only to infer $T_s(t)$. Define

$$J_4(t) = \int_0^t T_b^4(t') dt', \quad J_n(t) = \int_0^t T_b^n(t') dt', \quad Z_1 = f^2 \rho^{2(1-\mu)} (2 + \epsilon)(1 - \epsilon)C. \quad (35)$$

The zeroth-order compact ratio $u^{(0)}(t) = [T_s^{(0)}(t)/T_b(t)]^4$ is the unique root on $(0, 1)$ of

$$1 - u^{(0)}(t) = \left[\frac{Z_1 T_b^\beta(t) J_n(t)}{[2\sigma J_4(t)]^2} \right]^{1/2} [u^{(0)}(t)]^{(n+\beta)/8}. \quad (36)$$

For $k = 0, 1, \dots, K - 1$, compute

$$I_4^{(k)}(t) = \frac{\int_0^t u^{(k)}(t') T_b^4(t') dt'}{u^{(k)}(t) J_4(t)}, \quad I_n^{(k)}(t) = \frac{\int_0^t [u^{(k)}(t')]^{n/4} T_b^n(t') dt'}{[u^{(k)}(t)]^{n/4} J_n(t)}. \quad (37)$$

Then obtain $u^{(k+1)}(t)$ from the unique positive root

$$1 - I_4^{(k)}(t) u^{(k+1)}(t) = \left[\frac{Z_1 T_b^\beta(t) J_n(t)}{[2\sigma J_4(t)]^2} I_n^{(k)}(t) \right]^{1/2} [u^{(k+1)}(t)]^{(n+\beta)/8}, \quad (38)$$

solved by bisection or Brent on $0 < u^{(k+1)}(t) < 1/I_4^{(k)}(t)$. After K corrections,

$$T_s(t) \equiv T_s^{(K)}(t) = T_b(t)[u^{(K)}(t)]^{1/4}. \quad (39)$$

Thus $K = 0$ is the compact fluence model, $K = 1$ is the leading history correction, and increasing K approaches the full continuous HR boundary closure when the iteration converges.

Once $T_s(t)$ has been inferred, discard the bath-specific formulas and apply the standard surface-temperature HR results. Define

$$H(t) = T_s^n(t), \quad K_\epsilon = \frac{2 + \epsilon}{1 - \epsilon}. \quad (40)$$

The usual HR front formula is

$$x_F^2(t) = K_\epsilon C T_s^{-\beta}(t) \int_0^t T_s^n(t') dt'. \quad (41)$$

With

$$y = \frac{x}{x_F(t)}, \quad A(t) = \frac{x_F^2(t)}{CH^{1-\epsilon}(t)} \frac{\dot{H}(t)}{H(t)}, \quad (42)$$

the usual first-order HR profile is

$$T(x, t) = T_s(t) \left[(1 - y) \left(1 + \frac{\epsilon}{2} (1 - A(t)) y \right) \right]^{1/[n(1-\epsilon)]}, \quad (43)$$

for $0 \leq y \leq 1$, and $T(x, t) = 0$ for $y > 1$.

Numerically: interpolate $T_b(t)$, compute $J_4(t_i)$ and $J_n(t_i)$ by cumulative trapezoidal integration, solve (36), apply (37)–(38) through the chosen order K or until convergence, recover $T_s(t)$ from (39), and then evaluate the standard HR formulas (41)–(43).

The HR spatial expansion is unchanged: the retained front and profile are first order in ϵ , with neglected HR spatial terms of order $O(\epsilon^2)$. A finite K adds a boundary-history iteration error, which should be estimated from successive iterates. If the global iteration converges, the additional constant-ratio approximation is removed and the continuous HR boundary-energy closure is recovered. The remaining differences from an Appendix-A Cohen–Heizler calculation are then due to time quadrature, root-solver accuracy, and any difference in temporal discretization, rather than a new physical approximation.